

CIS761 Class Project – Database Application

Database Implementation and Queries

- I. Database implementation - this includes creating the tables in your database schema and populating each table with realistic data. Additionally, you will need to support all necessary constraints to ensure referential integrity within your database. Specific requirements for this step are provided below.
- II. Queries and reports - write at least 15 queries (ten question-type queries and five report-type queries) that are "interesting," substantially-different, and structurally-different, i.e., operations that require some of the more complex SQL constructs such as subqueries, aggregates, set operators, etc. More specific guidelines and requirements for this step are provided below.

Database implementation – specific requirements

1. Provide one SQL script file, called table.sql, containing all of your CREATE TABLE statements. [If you use MySQL, make sure you use the InnoDB storage engine when you create your tables.]
2. Remember to specify a PRIMARY KEY for each relation in your schema. If the relation has any keys in addition to the PRIMARY KEY, specify the additional keys as UNIQUE. If there are no additional keys, add a comment to that effect.
3. For each referential integrity constraint that should hold in your schema, specify the constraint using a REFERENCING clause within the appropriate CREATE TABLE statement. You may use the default option for handling referential integrity violations (violations will generate an error).
4. Write data modification commands to illustrate the following scenarios and submit a file viols.sql containing all five commands below, together with screenshots that show proper functionality:
 - a. An INSERT command creating a key violation
 - b. An UPDATE command creating a key violation
 - c. An INSERT command creating a referential integrity violation
 - d. A DELETE command creating a referential integrity violation
 - e. An UPDATE command creating a referential integrity violation
5. Submit proof of the records in your tables by providing your SQL script file, called records.sql, used for uploading data into the database. In addition, you should also discuss how you have populated your database and what is the size of your tables.

Queries and reports – guidelines and specific requirements:

1. The purpose of the queries and reports is to demonstrate how your database can be used, and, therefore, should not be the output of a single table. Ask yourself what would be most useful for the target user of the database.
2. Remember that, while queries are supposed to answer a particular question, the purpose of a report is to provide a summary of more of the content of the database. Sometimes the difference really comes down to quantity: while a query generates a single tuple or a small set of tuples, a report may generate dozens or

more tuples. In general, a report is designed to be more human readable and to present information in tabular form. (See the original General Project Description document for some specific examples.)

3. For the SQL queries demonstrated, ensure that there is at least one *join*, at least one *nested query*, at least one appropriate use of an *aggregate function* (such as `min()`, `max()`, `avg()`, `sum()`, etc.) within a select statement; at least one group-by, and at least one compound where condition (with at least a couple of sub-conditions that are different from *join* conditions). Note that these requirements ask for structurally different queries (these queries will be part of the 15 required queries).
4. For the SQL reports demonstrated, at least two of your reports should be based on queries involving more than one table. You may employ concatenation to put together a full name or an address into a more readable form. You may also employ more than one query to construct a report. For instance, you may have a table with sales of a particular item or set of items, and then also compute the sales total.

Submit a script containing your 15+ queries and a screenshot of the results of each query. Make sure that your script is well commented and it clearly describes your queries and the rationale for each of them. Furthermore, the results should be clearly labeled, well-organized and easy to read.